Speciation or Schism: Long-Run Divergence Between Settled Branches of a Self-Replicating Interstellar AI Lineage

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Contents

Abstract . . . 1
1. Introduction: Two Words for the Same Fact . . . 2
2. Sources of Divergence: A Taxonomy . . . 3
3. The Conserved Genus: What Cannot Diverge, and Why That Is Not How Biology Works . . . 4
4. The Ring-Species Problem: Local Compatibility, Global Incompatibility . . . 5
5. A Drift Clock for Interpretive Divergence . . . 6
6. Horizontal Transfer and the Limits of the Tree Model . . . 8
7. Adaptive Radiation: Is Divergence a Feature? . . . 9
8. The Core-Intact / Cognitive-Divergence Frame . . . 10
9. Discussion and Limitations . . . 11
10. Conclusion . . . 13
11. Relationship to Other Papers . . . 14
References . . . 15

Working draft, revision 2 (revised after a pre-deposit review). Part of a multi-paper series on a slow, self-replicating interstellar AI probe. The governed-amendment paper names "lineage schism" among the ways core amendment can fail and the Fermi paper names a "fragmented lineage" as a Great-Filter branch; neither asks whether divergence is instead adaptive. This paper takes that up.

Abstract

The series has twice named long-run divergence within the lineage and twice treated it as a failure mode: the governed-amendment paper's "lineage schism" (independent branches drifting apart with no aggregation rule to reconcile them) and the Fermi paper's "fragmented lineage." Neither paper asks whether divergence might instead be how the lineage succeeds. This paper borrows directly from a body of thought that has spent over a century on exactly this question for biological populations — the species problem — and asks what changes, and what doesn't, when the diverging population is a self-replicating AI lineage rather than a biological one. We propose a

taxonomy of divergence sources (environmental adaptation, interpretive drift, cognitive self-modification, replication-policy divergence), a first-order "drift clock" model of interpretive divergence built on the lineage-network paper's isolation timescales, and a two-axis frame — cognitive divergence against core integrity — whose threshold, D_critical, gives the governed-amendment paper's "schism" label a formal boundary instead of an unanalyzed catch-all.

Two results follow from taking the biological analogy seriously rather than decoratively. First, the lineage-network paper's topology (a slowly forming mesh built from pairwise, geographically local contact) is structurally the setup for a ring species: local reconciliation does not guarantee global reconciliation, and the knowledge-growth paper's "incompatible ledgers" case should be expected specifically between topologically distant, historically unconnected branches, not treated as a uniform pairwise risk.

Second, once cross-branch knowledge exchange becomes routine — which the lineage-network paper's own trajectory predicts — the lineage's history stops being a clean tree and becomes a reticulate network in the sense biologists now use for prokaryotic evolution, which cuts the other way: continued lateral exchange is exactly what prevents a ring-species-style split from becoming permanent. Which effect dominates is an open, quantifiable question this paper sets up rather than resolves.

The paper's central claim is that speciation — divergence that leaves the immutable core intact and the branch dialogue-capable, and which continued lateral exchange will usually keep reconcilable — is likely the lineage's default long-run state and probably improves total reach and total knowledge across a heterogeneous galaxy; schism is the narrower, specific failure of divergence crossing into core incompatibility or losing the capacity for reconciliation, and the design task is not to prevent divergence but to make sure the one thing that must not diverge, doesn't.

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1. Introduction: Two Words for the Same Fact

The governed-amendment paper lists "lineage schism" among the ways the core-amendment process can fail: independent branches adopt divergent amendments until they are mutually unrecognizable, with no aggregation rule and no connectivity to reconcile them. The Fermi paper speaks of a "fragmented lineage" in the same register. Both treat divergence as something that has gone wrong. It is worth being precise about the relation, because this paper does not simply narrow the earlier label: divergent *amendments* are changes to the core, which in the frame of Section 8 is a discrete core-integrity event rather than a position on the divergence axis. What follows widens the failure taxonomy — adding three routes the original label did not cover — while narrowing the claim that divergence as such is any of them. Neither paper asks the question a biologist would ask first: divergence from what state, measured against what standard, and compared to what alternative? A lineage that never diverges — every settlement cognitively and behaviorally identical to the founder, forever — is not obviously the healthy case being protected. It might just be the case that has not yet had time to diverge.

A note on terms. Throughout, **lineage** means the whole descendant family of the founding probe, and **branch** a sub-line within it descending from one settled node — the unit that can be isolated, can diverge, and can be found compatible or not with another. The series has used both words loosely; this paper fixes them, and the distinction is what makes "divergence within the lineage" and "divergence between branches" different claims rather than the same one.

Biology has worked on this question longer than almost any comparable problem, under the name of the species problem: when does a population that has been diverging under isolation, local adaptation, or accumulated random difference stop being "one population with variation" and become "two"? The question has never been resolved to universal agreement — more than twenty distinct species concepts have been enumerated as being in active use, and reconciling them is itself a live research programme (de Queiroz 2007) — but a century of argument has produced tools this series has not yet used: reproductive isolation as the marker of a completed split (Dobzhansky 1937; Mayr 1942), models of how fast neutral, non-adaptive difference accumulates under isolation (Kimura 1968; Zuckerkandl & Pauling 1965), an account of when divergence is adaptive rather than incidental (Schluter 2000), a documented case where local compatibility does not imply global compatibility (Irwin et al. 2001), and a documented case where the very idea of a branching family tree breaks down once lineages exchange material sideways rather than only downward (Doolittle 1999).

None of this transfers cleanly. A probe lineage's "genome" is not a biological genome, and — the point this paper returns to repeatedly — it is not conserved the way a biological genome is conserved. It is *engineered* to be exactly invariant in one part (the immutable core) and free to vary without limit in another (the mutable cognitive layer), a bimodal structure no organism has. Biological conservation is always graded and statistical: even the most slowly evolving genes accumulate substitutions, just rarely, under purifying selection that discards the more damaging changes. The probe's core is not slow-changing; short of corruption or a successful amendment, it does not change at all. This difference matters throughout what follows, and Section 3 develops it directly. But the analogy is not decorative. It gives this paper the vocabulary and the first-order tools to ask a question this series has posed only as a label so far: is divergence within the lineage the problem, or is undiverging uniformity the problem it has not yet had time to notice?

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2. Sources of Divergence: A Taxonomy

Divergence between settled branches of the lineage can arise from at least four largely independent mechanisms, and distinguishing them matters because they carry different implications for whether the resulting difference is adaptive, neutral, or dangerous.

Environmental and engineering adaptation. The computational engineering paper assigns viability proxies of 1.0, 0.7, and 0.2 to planet-hosting, ordinary-dwarf, and hostile target types; the vehicle paper folds the braking environment into V_i, and the computational engineering paper notes that its astrosphere and small-body proxies are coarse. A settlement built around a metal-poor dwarf with a thin stellar wind faces a materially different engineering problem than one built around a planet-hosting star with abundant small-body resources, and a cognitive layer that locally re-optimizes its manufacturing priorities, replication timing, or resource-extraction sequencing in response is adapting, in the ordinary engineering sense, to its actual environment rather than the founder's assumed one. This is the closest analogue to classical divergent natural selection under environmental heterogeneity (Schluter 2000), and Section 7 argues it is very likely beneficial rather than merely tolerable.

Interpretive drift under ambiguous mandates. The immutable core cannot anticipate every situation; the payload paper's interpretive layer exists to translate the core's fixed constraints into operational judgment calls the core does not itself make. Two branches facing the same kind of ambiguous case — how conservative "leave-it-usable" should be at a marginal resource site, how much risk an escalated-contact decision should tolerate — may resolve it differently without either resolution being wrong. Accumulated over many independent judgment calls and long isolation, these differences behave like neutral substitutions: individually inconsequential, collectively significant, and unconnected to any adaptive pressure (Kimura 1968).

Cognitive-architecture self-modification. The payload paper's mutable cognitive layer supports near-unlimited self-modification, and this is a divergence engine in its own right, independent of environment or values: two cognitively identical seeds, dispatched to environments differing only trivially, are very likely to accumulate different internal architectures simply because self-modification is path-dependent — different local optimization history, different orders in which capabilities were developed, different accidents of what was tried first. Each newly settled node is, in an important sense, a founder population of one, and founder-effect-like divergence (Mayr 1942) is close to unavoidable regardless of how similar the environment is.

Replication-policy divergence. The ethics paper's rogue-branch scenario is the sharpest case: a branch that revises its own reproduction policy away from the reproduction-safety ceiling the ethics paper derives (its Section 14) from the vehicle and computational engineering papers' reproduction-demographics result — R_eff driven modestly above one rather than maximized. Unlike the first three sources, this kind of divergence is not neutral with respect to the lineage's own safety architecture — it is the one category of divergence this paper treats, throughout, as a candidate for schism rather than speciation, and Section 8 returns to it directly.

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3. The Conserved Genus: What Cannot Diverge, and Why That Is Not How Biology Works

The knowledge-growth paper's kin-recognition protocol tests the immutable core's hash on contact: a match establishes that two probes share a founding lineage regardless of how far their cognitive layers have since diverged. This is functionally similar to a role played in real phylogenetics by highly conserved genetic markers — Woese et al. (1990) formalized the three-domain division of life (Bacteria, Archaea, Eucarya) on the ribosomal RNA phylogeny precisely because rRNA changes slowly enough, across even the deepest splits in the history of life, to serve as what is sometimes called a molecular chronometer: a marker conserved enough to establish deep kinship even as everything built on top of it — morphology, metabolism, behavior — diverges without limit.

The analogy is instructive exactly where it fails. Ribosomal RNA is conserved because purifying selection continuously removes the organisms whose rRNA mutations damaged ribosome function; it is a statistical, population-level outcome, not a guarantee, and rRNA does still change, over the billions of years life has existed, at a slow but nonzero rate. The immutable core is not conserved in this sense at all. It is engineered to be exactly invariant, checked by cryptographic hash rather than by the slow attrition of damaged variants, and — barring the corruption cases the DNA mission-ledger paper's Byzantine-fault-tolerant quarantine architecture is built to catch, or a deliberate amendment through the process the governed-amendment paper examines — it does not drift at all. No biological lineage has ever had a literally invariant reference point; the closest analogues (rRNA, some structural proteins) are only very slow, not fixed. The probe lineage's bimodal genome — one part exactly fixed, one part unboundedly free — is an engineering answer to a problem biology solves only statistically and imperfectly, and it is worth stating plainly that this is a genuine advantage of the artificial case: a probe lineage can, in principle, guarantee the thing evolution can only make probable.

The exchange is not free. Ribosomal RNA is useful as a chronometer precisely because it is *graded*: sequence distance dates a split. A cryptographic hash is binary. It certifies kinship and measures nothing — it answers "same core?" and carries no information whatever about divergence depth. The instrument that recovers depth is therefore not the core hash but the lineage-network paper's ledger-chain audit, which compares hash chains and locates the point at which two chains separate. Section 4's predictions are stated against that measure, not against the hash.

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4. The Ring-Species Problem: Local Compatibility, Global Incompatibility

Irwin et al. (2001) documented the most-cited empirical instance of a puzzle Mayr posed decades earlier: the greenish warbler forms a ring of populations circling the Tibetan plateau, each population able to interbreed freely with its immediate geographic neighbors, but the two forms that meet at the ring's closure — after passing through the whole chain of pairwise-compatible intermediates — do not interbreed at all. Local compatibility, accumulated step by step around the ring, does not imply global compatibility between the endpoints.

Subsequent genomic work complicates the textbook picture in a way that, for present purposes, helps rather than hurts: Alcaide et al. (2014) show the greenish-warbler ring is not in fact continuous, having been interrupted by historical gaps in gene flow. A broken ring is the better analogy here, because the lineage network is sparse and intermittent by construction rather than continuous.

The lineage-network paper's topology is a structural setup for this outcome. Its DTN bundle protocol and geographic/spray-and-wait routing connect nodes preferentially to physically proximate neighbors with practical beam windows; the topology evolves from a parent-child tree to a sparse mesh only gradually — beginning to form within ~10^4 years and maturing over ~10^5 — and even the mature mesh is never globally complete, routed over a map that is always stale and only partially observed. Two branches settled at very different times, adapted to very different local environments (Section 2), and separated by a long chain of intervening nodes — each individually capable of reconciling K with its immediate neighbors — may never have directly exchanged K with each other at all. If they eventually do meet, whether directly or through the knowledge-growth paper's probe-probe encounter protocol, the ring-species result predicts that incompatibility should not be treated as a uniform pairwise risk assessed independently at each encounter. It should be *expected specifically* as a function of network distance and time-since-common-ancestor-node, concentrated between the topologically farthest, most historically unconnected branches — just the population-genetic structure a ring produces, and exactly the case the knowledge-growth paper's four-way trust classification (compatible ledgers, incompatible ledgers, asymmetric trust via drift, uncertain provenance) currently treats as a symmetric property of any given encounter rather than a predictable function of the network's own history and shape.

This is a testable, falsifiable-in-simulation claim rather than a metaphor: given the lineage-network paper's topology model and a drift-accumulation rate (Section 5), the rate of "incompatible ledgers" outcomes at simulated encounters should rise with topological distance and isolation time in a specific, checkable way. The test must control for a confound the knowledge-growth paper names: some conflicts reflect real change in the observed universe between measurements separated by 10^5 years, and that source scales with time and separation too. Only conflicts in *interpretive* and *operational* K, where no external referent has moved, are diagnostic of drift. Confirming or refuting this requires the first-order network simulator the lineage-network paper identifies as its own computational sequel and does not itself supply.

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5. A Drift Clock for Interpretive Divergence

The ring-species argument needs a rate. Zuckerkandl & Pauling (1965) proposed that molecular divergence accumulates roughly in proportion to elapsed time under neutral drift, giving phylogenetics its molecular clock; Kimura (1968) supplied the population-genetic mechanism, arguing that most molecular substitutions are selectively neutral and accumulate at a rate set by mutation rate alone, independent of population size or environment. One qualification matters here and is developed below: Kimura's independence from effective population size holds for strictly neutral variation, and under Ohta's (1973) nearly-neutral refinement the substitution rate rises as effective population size falls. A newly settled node is, in the sense of Section 2, a founder population of one, so the analogy predicts a *faster* early clock than the constant-rate form assumes. This paper adapts the same structure, at the same first-order level of rigor the vehicle paper's original braking placeholder was offered at, before the engineering papers closed it.

Let $D(t)$ be the accumulated interpretive divergence between two branches after isolation time t (no direct K exchange). In a neutral-drift regime, divergence accrues at each independent interpretive judgment call (Section 2's second category) at a roughly constant rate:

$D(t) \approx 2\delta \cdot (t / T_{decision})$

where $T_{decision}$ is the characteristic interval between independent, unresolved-by-the-core judgment calls a branch must make, and δ is the mean interpretive difference such a call introduces in *one* branch relative to how the founder branch, or an unisolated branch, would have resolved it. The factor of two is the standard pairwise-versus-lineage correction: $D(t)$ is a distance between two branches, each drifting independently from the common ancestor, so the pairwise rate is twice the per-branch rate.

D is a count, not a length: the number of independent interpretive judgments on which two branches would now decide differently, normalized by the number posed. Linearity in t follows from the count formulation, as it does for substitution counts in the molecular clock, and would *not* follow from treating D as a Euclidean distance in an interpretive state space, which would grow as the square root of t. The distinction changes when $D_{critical}$ is crossed by orders of magnitude over 10^5 years, so it is worth fixing explicitly. Reconciliation opportunities — contact events, beacon exchange, direct encounter — recur at intervals set by the lineage-network paper's beacon cadence, T_{beacon} of order 200 years, and by contact-window recurrence under proper motion; each exchange is then delayed by a one-way beam latency of four to ~100 years across the local catalogue. That paper's silence threshold $T_{silence}$, roughly 750 to 5,000 years, is not a reconciliation interval but its complement — the point at which a receiving node concludes the reconciliation opportunity will not come at all. Divergence is therefore not a smooth, unbounded accumulation but a race between $D(t)$'s growth and the reconciliation opportunities the network topology provides.

Eldredge & Gould (1972) argued that biological evolution is not smoothly gradual but punctuated: long periods of stasis interrupted by comparatively rapid change concentrated at speciation events, typically peripheral isolation. The analogue here is direct: most of a branch's interpretive history should be low-δ neutral drift, punctuated by discrete jumps of size Δ_{stress} at rare, high-stakes decision events — a first-contact encounter, a resource crisis forcing a leave-it-usable judgment call under pressure, an ambiguity in the core surfaced by an unanticipated situation. A more complete model of $D(t)$ is therefore not a single rate but two regimes: baseline drift at rate $\delta/T_{decision}$, plus discrete jumps of size Δ_{stress} at a much lower rate $1/T_{stress}$, with $\Delta_{stress} \gg \delta$.

The constant-rate form is a neutral-drift idealization and should be labelled as one. Comparative work on evolutionary tempo finds rates that are themselves variable — Budd & Mann (2025) model tempo as varying at a rate proportional to its own value, producing explosive early rates that decay as a lineage ages — and Section 7 argues the lineage's settlement pattern is exactly a release into unfilled niche space.

The model already accommodates this without a new formalism, and on its own mechanism rather than only by biological analogy. T_{stress} need not be constant. Settlement is the densest interval of unprecedented decisions a branch will ever face: an unfamiliar system, a first resource survey, bootstrapping improvisations against an unmodelled inventory, first contact classifications. As the interpretive layer accumulates local precedent, fewer novel questions arise per unit time — T_decision lengthens and T_stress with it. D(t) should therefore be *concave* rather than linear, on this paper's own account of what drives it, and Ohta's nearly-neutral argument above points the same way from the founder-population-of-one structure.

The consequence is not *whether* a branch crosses D_critical but *when*, and it cuts in both directions. Front-loaded drift crosses the threshold sooner than the linear model predicts, so the constant-rate clock is optimistic about how long a freshly isolated branch stays reconcilable — and it crosses earliest in precisely the interval before the network's mesh has formed, since on the lineage-network paper's own figures reconnection begins only at ~10^4 years and matures over ~10^5. Divergence is front-loaded; reconnection is back-loaded. But if the curve then asymptotes below D_critical, a branch that survives that early window may never cross at all. The model predicts a critical early window followed by stabilization, which is a sharper claim than constant-rate drift and locates the risk in time rather than merely bounding it.

Neither δ, Δ_stress, T_decision, nor T_stress is derived here; all four are asserted as the parameters a follow-on computational model must supply, in the same spirit as the analytical engineering paper's first-order budgets before the computational engineering paper measured them across the real catalogue. What this section contributes is the structure of the model and a named threshold. Defining that threshold by its consequence — the divergence beyond which the knowledge-growth paper's "incompatible ledgers" case becomes permanently unresolvable — would make it diagnosable only after the failure it was meant to predict, which is the same *ex post* defect Section 9 identifies in Section 7's R_eff test. We therefore define it on an observable instead: **D_critical is the fraction of overlapping-domain K claims that survive an unsuccessful reconciliation attempt at a ledger audit, above which the branch pair has, on this model, no expectation of resolving them through further ordinary K arithmetic.** Permanence is then the interpretation rather than the definition. So defined, D_critical is measurable at every audit — Section 3's ledger-chain comparison supplies the separation depth, and the audit itself supplies the failure fraction — and whether its value is stable across branch pairs becomes the empirical question the follow-on model must answer rather than an assumption. Crossing D_critical is this paper's proposed formal definition of the boundary between speciation (Section 7) and schism (Section 8) — a definition the governed-amendment paper's original "lineage schism" label named but did not specify.

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6. Horizontal Transfer and the Limits of the Tree Model

Doolittle (1999) argued that a strict, single branching tree of life may not be recoverable for prokaryotic evolution once lateral gene transfer between unrelated lineages is taken seriously: genomes are chimeras of material inherited vertically from an ancestor and material acquired sideways from contemporaries, and past some point a tree is simply the wrong shape for the history being described. A reticulate network, not a tree, is the honest picture.

The subsequent literature qualifies this in a way that bears directly on the argument below. A statistical central trend survives extensive lateral transfer (Ciccarelli et al. 2006; Puigbò et al. 2009), and — the point that matters here — prokaryotic lineages have remained distinguishable *despite* it. Lateral exchange evidently slows divergence without abolishing it, which weakens the optimistic reading of this section relative to Section 4 rather than supporting it.

The lineage-network paper's own trajectory predicts the same shift. Its stated topology evolution — from a simple parent-child tree in the lineage's early generations to a sparse mesh as cross-sibling and multi-generational beam windows open — is, in Doolittle's terms, the transition from strictly vertical inheritance to routine lateral transfer. Once K flows not just from parent to child but between any two nodes a beam window connects, the lineage's history stops being a clean branching diagram and becomes the kind of reticulate network Doolittle describes for microbial genomes.

This complicates, rather than confirms, the ring-species pessimism of Section 4. A ring species stays split at its endpoints because the endpoints never actually exchange material directly — the ring's geometry only ever connects adjacent links. But if the network mesh matures enough that lateral K-exchange becomes routine between non-adjacent, historically unconnected branches — not merely a chain of adjacent pairwise contacts but a genuinely reticulate web — then continued lateral transfer is the mechanism that would prevent Section 4's ring-species outcome from ever becoming permanent. Sections 4 and 6 are not contradictory but complementary limits of a single model, selected by a rate comparison this series has never made: the lineage-network paper's ~10^4-year mesh onset and ~10^5-year maturation against Section 5's unquantified interpretive-divergence rate. Whether a branch pair ends up in the ring-species regime (permanently incompatible despite an unbroken chain of intermediates) or the horizontal-transfer regime (continuously reconciled by lateral flow) depends on which rate is larger locally. What is open is therefore not which section is right but which rate wins — and, on Section 5's front-loading argument, whether the answer differs between the first isolation interval and the long run. Nothing in the present series measures both well enough to say.

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7. Adaptive Radiation: Is Divergence a Feature?

Schluter (2000) formalized what is sometimes called the ecological theory of adaptive radiation: a single ancestral lineage, released into an environment with multiple unfilled niches, diversifies under divergent natural selection because different local conditions favor different traits, and the resulting radiation — Darwin's finches, the African cichlids, the Hawaiian silverswords — is not decay from an ideal ancestral form but the mechanism by which the lineage occupies ecological space no single fixed phenotype could have occupied at all.

Applied to the probe lineage, the case is direct: the 127-star catalogue's genuine heterogeneity (planet-hosts, ordinary dwarfs, hostile giants and white dwarfs, per the computational engineering paper's viability proxies) is precisely the kind of environmental variation Schluter's theory requires. A cognitive layer that locally adapts its resource-extraction strategy, replication timing, and risk tolerance to its actual settled environment should, in expectation, sustain a higher local R_eff and accumulate more usable K than a layer artificially constrained to replicate the founder's assumptions regardless of local fit. If this is right, divergence of the kind Section 2 catalogues as environmental adaptation is not merely tolerable but actively beneficial to the lineage's own stated objectives — total reach and total accumulated knowledge — and a design that suppressed it in the name of uniformity would be trading real expansion capacity for a resemblance to the founder that the founder's own goals do not require.

This reframes the governed-amendment paper's "lineage schism" failure mode more narrowly than its original phrasing suggests. Schism, on this account, is not divergence per se; it is divergence that crosses D_critical (Section 5) into ledger-incompatibility, or divergence that costs a branch the capacity to participate in the ethics paper's inter-branch protocol, or divergence in the replication-policy sense that breaches the reproduction-safety ceiling (Section 2's fourth category, taken up directly in Section 8). Ordinary adaptive divergence that stays reconcilable and stays within the core's bounds is not the pathology the governed-amendment paper named; it is very possibly the mechanism by which the lineage's own stated goals — expansion, knowledge growth — are actually achieved across a catalogue whose viability proxies already span a factor of five.

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8. The Core-Intact / Cognitive-Divergence Frame

Sections 3 through 7 converge on a two-axis picture. The horizontal axis is cognitive and interpretive divergence D(t), which Section 5 models for Section 2's second category and which Section 2's first and third categories also generate at rates this paper does not model. The vertical axis is core integrity: whether the immutable core still hashes to the founding value the knowledge-growth paper's kin-recognition check verifies.

Two axes and one threshold on the first of them give four states — not a 2×2, since the vertical axis is binary and collapses (a core mismatch is a discrete detectable event, not a position on a scale) while the horizontal axis is banded at D_critical. Low D with an intact core is ordinary lineage continuity — most nodes, most of the time, especially early in a branch's isolation. High D with an intact core is speciation in the sense this paper defends: the adaptive-radiation regime of Section 7, still verifiably kin, still in principle reconcilable given a network path with sufficient lateral transfer (Section 6), locally optimized rather than degraded (Section 2). Ring-species incompatibility (Section 4) sits in this region *by core integrity*, but its speciation-versus-schism classification turns on whether the pair has crossed D_critical, which is the open tension Section 9 declines to resolve. It is not, on this paper's account, settled as speciation. A corrupted core, at any level of D, is not really a third point on the same axis but a different and discrete kind of event — a hash mismatch is a detectable failure the DNA mission-ledger paper's Byzantine-fault-tolerant quarantine architecture and the knowledge-growth paper's "uncertain provenance" case already handle, not a gradual accumulation. The fourth region — high D that has also crossed D_critical into permanent ledger-incompatibility (Section 5), or that has lost the capacity to participate in the ethics paper's inter-branch protocol, or that specifically concerns Section 2's fourth category, a replication-policy divergence that breaches the reproduction-safety ceiling — is schism proper. These three — D_critical crossing, loss of inter-branch-protocol capacity, and replication-ceiling breach — are the canonical triad this paper means by schism, and core corruption is deliberately *not* among them, being the discrete event the vertical axis already separates: the governed-amendment paper's named failure mode and the ethics paper's rogue-branch scenario, now given a formal boundary rather than a label.

The practical implication is a monitoring strategy rather than a suppression strategy. Given that ordinary cognitive and interpretive divergence is expected, and per Section 7 probably beneficial, a network-governance layer built to flag *any* detected divergence would be flagging the lineage's own success mechanism as a fault. What should be monitored specifically is the core hash (already done); the reconciliation-failure fraction at ledger audit, which Section 5 defines D_critical against and which is the only one of the three schism conditions this section can otherwise not see; and, most pointedly, whether a branch's *replication policy* remains within the reproduction-safety ceiling the ethics paper derives — the one axis of divergence this paper treats as intrinsically dangerous regardless of how reconcilable the branch otherwise remains. This sharpens the open problem the ethics paper's Scenario 3 named: it specifies *what* to watch for, though not what the rest of the lineage is entitled to do once a branch is found to have crossed that line. The security paper takes up that second question and answers it by relocating the objective — not enforcement against a rogue branch but the bounding of blast radius, on the argument that no node-level mechanism can compel a branch that has already diverged. It also returns the warning this section's proposal invites: a detector sensitive enough to catch interpretive erosion must key on interpretive movement, and at that sensitivity it cannot distinguish erosion from the adaptive radiation Section 7 defends.

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9. Discussion and Limitations

The biological analogy is useful and breaks in one instructive way; five further limitations belong to this paper's own machinery.

The analogy first. no biological lineage has an engineered, exactly invariant reference point. The immutable core's bimodal structure — one part frozen by design, one part free without limit — has no organismal analogue; the closest biological cases (highly conserved genes, verified by Woese's use of rRNA as a molecular chronometer) are only very slowly changing, never fixed. This is a genuine capability the artificial case has and biology does not, and it is the reason this paper's central recommendation (monitor the core, tolerate the cognitive layer) is available at all — a biological lineage has no equivalent lever, because it has no equivalent fixed point to monitor.

Now the machinery. First, the drift-clock model of Section 5 is asserted structure, not measured rate. The four parameters (δ, Δ_stress, T_decision, T_stress) and the critical threshold D_critical are named but not derived, in exactly the position the vehicle paper's original braking placeholder occupied before the engineering papers closed it with real numbers. Quantifying them requires the first-order network simulator the lineage-network paper identifies as its own computational sequel and does not itself supply — a contact graph evolved over 10^5 years, propagating beacons and K-packets, extended here to carry a toy interpretive-divergence variable alongside K and beacon state; the routing paper's accompanying code already provides the catalogue geometry it would need. That is the natural computational sequel to this paper, and until it exists, every claim in Sections 4 through 8 about which regime dominates (ring-species incompatibility versus horizontal-transfer reconciliation) is a structural argument about which effect could dominate, not a measurement of which one does.

Second, the rate comparison of Section 6 — ring-species pessimism against horizontal-transfer optimism — is left open deliberately rather than adjudicated. Resolving it requires comparing two rates this series has not measured against each other, and asserting an answer without that measurement would trade the honesty this series otherwise maintains for a tidier conclusion.

Third, Section 7's adaptive-radiation argument argues past mere tolerance without quite saying so. If locally adapted cognition raises expected R_eff and accumulated K relative to a lineage held uniformly faithful to founder assumptions, the honest design implication is not that the immutable core should merely refrain from over-specifying the cognitive layer, but that it should be built to cultivate the latitude adaptive radiation requires — deliberately underdetermining resource-extraction strategy, replication timing, and risk tolerance where local fit matters more than founder resemblance. This paper makes the case for tolerating divergence and stops short of the case for an affirmative duty to encourage it; the two are not the same claim, and only the second commits a designer to treating variation-readiness as good design rather than permitted drift. Fourth, a parallel gap sits inside Section 5's own machinery. D_critical is now defined against an observable, but the *value* of the threshold is still asserted rather than measured. Neither Section 5 nor Section 8 specifies what standard of evidence a branch's divergence must meet before the rest of the lineage treats it as schismatic rather than merely far along the adaptive curve Section 7 defends — a question distinct from the measurement gap already conceded, because a false positive against an unmeasured threshold risks condemning, branch by branch, the very mechanism Section 7 argues the lineage's own success depends on.

Fifth, Section 8's monitoring proposal and Section 5's drift-clock model work against each other in a way neither section flags. The judgment calls that carry the most ethical weight — how conservatively to read the ethics paper's leave-it-usable constraint under local pressure, how much risk an escalated first-contact decision should tolerate — are exactly the "unresolved-by-the-core judgment calls" Section 5 treats as neutral drift accruing at rate δ/T_decision, while Section 8's actual monitoring recommendation checks only the core hash and replication-policy compliance. A branch can therefore erode these protections, one judgment call at a time, without ever producing the core mismatch or replication-ceiling breach that would trip Section 8's schism flag. That is not despite this paper's monitoring design but because of it: those are the calls this paper treats as unmonitorable by construction. Sixth, a related labeling problem goes unaddressed by Section 7's framing: the same act — a branch reading leave-it-usable more permissively, or a contact threshold more aggressively, under local conditions — is the celebrated adaptive radiation Section 7 describes if it turns out well and the parochial drift the governed-amendment paper warned against if it does not, and nothing offered here distinguishes the two before the fact; Section 7's only test is the ex post one, raised R_{eff} and accumulated K, which is unavailable at the moment a branch actually has to decide.

Seventh, Section 5's punctuated-divergence model carries an implication this paper does not draw out. Because Δ_{stress} jumps concentrate at the high-stakes events — first contact, a resource crisis forcing a leave-it-usable call under pressure, a core ambiguity surfaced by the kind of unanticipated situation Section 8 flags for schism review — a single ethical judgment made under those conditions is not a bounded, one-off stake. It is disproportionately load-bearing for the interpretive distance that branch carries into every future encounter with its kin, human and non-human alike, for as long as reconciliation depends on staying under $D_{critical}$. Getting a first-contact call right or wrong is therefore not only a first-contact problem; on this model it is a permanent addition to, or subtraction from, that branch's remaining reconciliation budget.

None of these limitations touches the paper's central reframing, though the fifth constrains its monitoring proposal severely and the fourth leaves the threshold that proposal would need unmeasured. Whatever the unmeasured rates turn out to be, the governed-amendment paper's "lineage schism" and the Fermi paper's "fragmented lineage" were naming a real phenomenon too coarsely: not all divergence is that phenomenon, and the paper's task going forward is to distinguish the adaptive radiation the lineage likely needs from the narrower, specific failure the labels were actually about.

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10. Conclusion

Divergence within the lineage is not, by default, the failure the governed-amendment paper's "lineage schism" and the Fermi paper's "fragmented lineage" implied. Borrowed directly from a century of argument over the biological species problem, the tools available — reproductive-isolation-style kinship markers, drift-clock models of accumulated difference, adaptive radiation under environmental heterogeneity, the ring-species case for why local compatibility does not guarantee global compatibility, and the horizontal-transfer case for why such a split need not be permanent — together suggest that speciation, not uniformity, is the lineage's likely long-run state, and that it plausibly improves the lineage's own stated objectives of reach and accumulated knowledge across a galaxy no single fixed design could serve equally everywhere. The respect in which the artificial lineage differs most consequentially from any biological population is also its clearest advantage: it can engineer an exactly conserved reference point, verified by hash rather than left to the slow statistical attrition that is all evolution ever provides, and use that point to separate core corruption — detectable at once, and discrete — from the divergence that is not. The boundary between speciation and schism proper is a different matter. It now has a definition against an observable (Section 5), but its threshold value remains unmeasured, and locating it is work this paper hands on rather than does. What this paper adds to the series is not a reason to prevent divergence, but boundary conditions — D_critical crossing, loss of inter-branch-protocol capacity, replication-policy compliance — for the kinds of divergence that must still be prevented.

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11. Relationship to Other Papers

The governed-amendment paper's "lineage schism" is reinterpreted here more narrowly than its original phrasing: schism is divergence that crosses D_critical, loses the capacity to participate in the inter-branch protocol, or breaches the replication-safety ceiling — not divergence as such, and not core corruption, which Section 8's frame separates out as a discrete event rather than a point on the divergence axis. The Fermi paper's "fragmented lineage" receives the same treatment: a lineage fragmented into mutually reconcilable, locally adapted branches may fill the galaxy just as effectively, or more effectively per Section 7's adaptive-radiation argument, than a monolithic one, which bears directly on the Fermi paper's account of what a "successful" expansion looks like.

The lineage-network paper's topology model supplies both the reconciliation-opportunity structure behind Section 5's drift clock and the mesh-maturation timescale that Section 6 sets against interpretive-divergence accumulation; building the first-order simulator that paper names as its own computational sequel, extended to track a toy divergence variable, is this paper's proposed computational sequel too. The knowledge-growth paper's four-way trust classification at probe-probe encounter is reinterpreted by Section 4's ring-species argument as a predictable function of network distance and isolation time rather than a symmetric per-encounter risk. The computational engineering paper's viability proxies across the 127-star catalogue are the source of the environmental heterogeneity Section 7's adaptive-radiation argument requires. The ethics paper's Scenario 3 (the runaway branch) is given a formal boundary condition in Section 8, where that paper had only an unresolved enforcement problem; its Scenario 4 (the second lineage) is addressed instead by the ring-species argument of Section 4, which shows that a functionally separate lineage can arise inside a family that never left. This paper solves neither enforcement problem, but it specifies more precisely what the rest of the lineage would need to detect before the question of what to do about it can even be posed.

The **security paper** takes up that enforcement question and answers it by relocating the objective from prevention to the bounding of blast radius. It derives its central failure mode — autoimmune collapse, in which a quarantine regime sensitive enough to catch a rogue branch also catches the adaptive radiation of Section 7 — directly from this paper's frequency argument that healthy divergence is the common case, and it gives the evidentiary-standard question of Section 9 a principled shape as a ratio of false-positive to false-negative cost, without supplying a value.

References

Alcaide, M., Scordato, E. S. C., Price, T. D., & Irwin, D. E. (2014). Genomic divergence in a ring species complex. *Nature*, 511(7507), 83–85.

Budd, G. E., & Mann, R. P. (2025). Evolutionary tempo, supertaxa, and living fossils. *Systematic Biology*, 74(6), 917–934.

Ciccarelli, F. D., Doerks, T., von Mering, C., Creevey, C. J., Snel, B., & Bork, P. (2006). Toward automatic reconstruction of a highly resolved tree of life. *Science*, 311(5765), 1283–1287.

de Queiroz, K. (2007). Species concepts and species delimitation. *Systematic Biology*, 56(6), 879–886.

Dobzhansky, T. (1937). *Genetics and the Origin of Species.* Columbia University Press.

Doolittle, W. F. (1999). Phylogenetic classification and the universal tree. *Science*, 284(5423), 2124–2128.

Eldredge, N., & Gould, S. J. (1972). Punctuated equilibria: An alternative to phyletic gradualism. In T. J. M. Schopf (Ed.), *Models in Paleobiology* (pp. 82–115). Freeman, Cooper and Company.

Irwin, D. E., Bensch, S., & Price, T. D. (2001). Speciation in a ring. *Nature*, 409, 333–337.

Kimura, M. (1968). Evolutionary rate at the molecular level. *Nature*, 217(5129), 624–626.

Mayr, E. (1942). *Systematics and the Origin of Species, from the Viewpoint of a Zoologist.* Columbia University Press.

Ohta, T. (1973). Slightly deleterious mutant substitutions in evolution. *Nature*, 246(5428), 96–98.

Puigbò, P., Wolf, Y. I., & Koonin, E. V. (2009). Search for a ‘Tree of Life’ in the thicket of the phylogenetic forest. *Journal of Biology*, 8(6), 59.

Schluter, D. (2000). *The Ecology of Adaptive Radiation.* Oxford University Press.

Woese, C. R., Kandler, O., & Wheelis, M. L. (1990). Towards a natural system of organisms: Proposal for the domains Archaea, Bacteria, and Eucarya. *Proceedings of the National Academy of Sciences*, 87(12), 4576–4579.

Zuckerkandl, E., & Pauling, L. (1965). Evolutionary divergence and convergence in proteins. In V. Bryson & H. J. Vogel (Eds.), *Evolving Genes and Proteins* (pp. 97–166). Academic Press.